Piyal Chakraborty

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Chattogram, Bangladesh
LinkedIn | GitHub

SUMMARY

Bachelor of Science student in Electronics and Telecommunication Engineering at CUET, passionate about integrating Machine Learning with VLSI — particularly for semiconductor manufacturing, wafer map defect classification, and hardware-aware AI. Eager to apply deep learning and data science to real-world chip design and fabrication challenges, collaborate on cross-disciplinary projects, and drive impactful solutions in the semiconductor industry.

SKILL SUMMARY

Languages: Python, C, C++, HTML, React.js, Tailwind CSS, Node.js
Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib,TensorFlow, PyTorch
Tools: MS Word, Excel, PowerPoint
Platforms: PyCharm, Jupyter Notebook, VS Code, Kaggle

EDUCATION

B.Sc in Electronics and Telecommunication Engineering (ETE)
Chittagong University of Engineering & Technology (CUET) 01/2023 – Present
CGPA: 3.85 (Till 6th Semester)
Relevant Coursework: Signals & Systems, Control Systems, Data Structures & Algorithms, Digital Signal Processing, Digital Communication, CCN, Microprocessor & Microcontroller, VLSI, AIC, DIC etc.

Higher Secondary Certificate (HSC)
Govt. Hazi Mohammad Mohsin College 01/2019 – 12/2021
GPA: 5.00

PUBLICATION

A Parallel CNN-ViT Feature Extraction Framework with Bayesian-Optimized SVM for Semiconductor Wafer Map Defect Classification
Presented at the 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025).
Published in Springer Lecture Notes in Systems.
Springer Link

PROJECTS

EchoSignRealTime 2025 – Present
Sign language recognition project using Deep Learning and Computer Vision techniques. Focused on gesture classification and real-time prediction.
GitHub Repository

Deep Learning Projects 03/2025 – Present
Repository covering deep learning fundamentals and practical implementations using PyTorch.
GitHub Link

Machine Learning Projects 08/2024 – Present
Implemented ML algorithms, preprocessing techniques, and model evaluation using Scikit-learn, TensorFlow, and PyTorch.
GitHub Link

Pandas Projects – Data Analysis & Manipulation 07/2024 – Present
Projects demonstrating data cleaning, analysis, and visualization using Pandas.
GitHub Link

Python Projects 11/2023 – Present
Collection of core Python projects covering problem-solving and practical applications.
GitHub Link

Library-Management-System 09/2023 – 10/2023
a full-stack Library Management System built to streamline library operations such as managing books, users, and records efficiently with React, Tailwind CSS Backend: Node.js, Express.js, Python & MongoDB
GitHub Link

CERTIFICATIONS

Finalist – SciBlitz 1.0 (3MT Thesis Competition)
One of the top 40 finalists in the first-ever 3MT Thesis Competition of Bangladesh, organized by IEEE Student Branch, CUET.
Certificate Link

1st Author – A Parallel CNN-ViT Feature Extraction Framework with Bayesian-Optimized SVM for Semiconductor Wafer Map Defect Classification
Presented at the 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025).
Certificate Link

EXTRA CO-CURRICULAR ACTIVITIES

Executive Member, IEEE CS CUET Student Branch Chapter
Executive Member, IEEE CUET Student Branch
General Member, Joyoddhony CUET